

Automated Detection of Skin Tone Diversity in Visual Marketing Communication

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Abstract

Companies invest heavily in diversity, equity, and inclusion efforts. Specifically, the representation of people in visual marketing communication is often considered a manifestation of diversity policies. We propose a standard framework built on machine learning to create novel measures quantifying skin tone dynamics. We first use the Swin Transformer to extract skin pixels from images. Next, the K-means algorithm is deployed to classify skin tone components from the extracted skin pixels, accounting for multiple people with distinct skin colors in an image. Using images posted by 34 fashion brands on Instagram and Twitter, we demonstrate a useful application of the tool. The results highlight that, in the past two years, the fashion industry has slightly increased its diversity, represented by the increased variety of skin tones of people included in social media posts. Our method allows for automated detection of objective measures of skin-tone diversity in visual marketing communications.

Keywords: diversity, skin tone, machine learning, image mining, social media

1. Introduction

Companies are under tremendous pressure to manage their diversity, equity, and inclusion (DEI) efforts. Internally, the policies are proven to be vital in creating and maintaining a successful workplace. Externally, consumers are more likely to engage with a brand that looks diverse (GYNN, 2020). A research report conducted by Adobe in 2019 revealed that for 61% of Americans, diversity in marketing is crucial to their purchase decisions (Adobe, 2019). Consequently, studies from McKinsey & Company and The Wall Street Journal show that companies with different

levels of diversity perform heterogeneously in terms of operating results, innovation, and profits (Watkins, 2021). However, most marketers agreed that more can still be done, especially regarding how diversity is portrayed in marketing contents (Molenaar, 2021).

One particular area of marketing for DEI is representation. While companies actively demonstrate their DEI commitment through marketing communications, the communication content can also be regarded as informational cues representing their DEI standpoint. For example, brands post curated visual content to promote their products, shape consumer perceptions, and increase awareness on social media. The choices of people, their sizes and skin tones, and the diversity of such can strategically signal the brands' inclusivity and legitimacy to consumers. Equally, consumers can interpret the diversity signals represented in the content. While genuinely diverse and inclusive content that resonates consciously and subconsciously with audiences and stakeholders requires far more than an image (GYNN, 2020), our study responds to the call that there is a pressing need to study media representations of diversity (Arsel et al., 2022).

Variety in skin tones in the visual presentations has been considered a reflection of diversity (Sweeney & Whaley, 2019). Some brands could argue this is to strategically deploy people with desired skin tones to advertise their products for targeting specific marketing segments. The representation of skin tones in brands' visual communications could vary across industries and sectors, and even over time. Prior studies have researched color representation in the beauty (Frisby et al., 2019) and fashion industry (Butkowski et al., 2022), and confirmed an overall tendency toward lighter-skinned people. Therefore, it is crucial to be

aware of how brands and their competitors position the diversity signaling before taking actions in response to consumer needs or social forces. This can be an emotionally loaded topic, so objective measures are warranted. Currently, there is a lack of consistent understanding of how skin tone diversity can objectively be assessed. This, in turn, can be problematic when dealing with large unstructured datasets.

Analyzing unstructured media data (i.e., text, images, and videos) is challenging for marketing researchers because they usually have non-numeric values, multiple facets, and concurrent representations (Balducci & Marinova, 2018). Existing studies on human skin tone in images used manual coders, which is time-consuming and cumbersome (Mitchell, 2020). Additionally, it is economically and practically unfeasible to analyze skin tone in video messages. To conquer the technical challenges, this study proposes a standard framework based on machine learning (ML), focusing on automatically analyzing the skin tone of fashion people present in unstructured images or videos. We ask readers to interpret the results cautiously because skin tone, unequal to race, can manifest regardless of the race, and significant variation can exist within individuals with the same race (Hermosilla et al., 2018)

We use a three-pronged approach to quantify diversity. First, we train a Swin Transformer (Liu et al., 2021), a supervised ML model, to detect the skin of people portrayed in visual content. Second, we apply K-means, an unsupervised ML model, to classify the extracted skin pixels into different clusters to account for various skin tone components in a single image, such as multiple people with varying skin colors. Third, we operationalize the skin tone measure of an image as the brightness of the average skin tone, which reflects how light or dark the color of the human skin is. Furthermore, we propose novel skin tone diversity measures, including both categorical and continuous approaches. For the categorical skin tone metric, we adopt and extend the skin tone thresholds from Mitchell, 2020 and define five skin tone labels: very light, light, medium, dark, and very dark. For the continuous skin tone metric, we use the variance and mean of skin tone brightness. The variance and mean explaining the variation of skin tone representations reflect the long-term changes in brand communication strategies.

To summarize, we propose a standard framework and demonstrate how we can objectively evaluate and benchmark skin tone diversity in visual marketing communications. Substantively, we contribute to the marketing DEI literature by shedding new light on understanding the skin tone dynamics in marketing communications. More importantly, our research makes

an important methodological contribution to the visual marketing literature. To the best of our knowledge, this is the first framework that is capable of automatically detecting skin tone diversity in online images and videos. Our framework is important and timely, because it allows companies to analyze and be aware of their own and the industry-standard in DEI signaling. This method helps overcome the challenge of measuring diversity in large-scale unstructured data. It will be available as an online tool, which will enable researchers to study diversity at scale or easily incorporate diversity metrics as part of their methodological framework. Overall, this study presents an important next step in the research stream of DEI in marketing.

2. Background

Diversity can be broadly defined as "the existence of differences or variations within a group on certain demographic variables" (Auh & Menguc, 2005). It is, however, contextual because the perception of such can vary depending on the focal variables and the individuals' own diversity background (Li et al., 2021). Under the principles of DEI marketing initiatives, diversity can be about the presence of differences that may include ethnicity, gender, age, religion, and sexual orientation. One way for companies to demonstrate their priorities is through visual representations of these characteristics (Walters, 2018). In particular, skin color representation, which is often associated with colorism, has presented a particularly harmful practice in the marketplace (Arsel et al., 2022). Coloristic preferences operate over the skin tone of humans, where certain skin tones might be preferred over others in different contexts (Hunter, 2007). Existing studies consistently found that human skin tone in brand communications impacts consumer behaviors and subsequent business strategies.

For example, Meyers, 2011 examined the skin tone in targeted marketing and found that Black participants that were stronger in ethnic identification felt more positively towards darker-skinned people than those Black participants that were weak in ethnic identification. The influence of skin tone runs beyond ethnicity identification. A study in service industries found that skin tone identity directly influenced customer evaluations among Black, Hispanic, and White customers (Cowart & Lehnert, 2018). The general preference for the lighter skin tone may be culturally-rooted and not necessarily ethnically motivated. A prior study investigated Chinese colorism in Hollywood castings and observed that Chinese society's aesthetic preference for lighter skin could be

linked to the more frequent casting of pale-skinned stars in films targeting the Chinese market (Hermosilla et al., 2018).

Prior researchers consistently confirmed that skin tone is a non-negligible factor in understanding consumer behaviors. However, there is a lack of studies investigating the diversity of skin tone representations in marketing communications. A handful of prior studies used content analysis and focused more on the representations of specific skin tones (Frisby et al., 2019). The studies typically used manual coders to quantify the skin tone for marketing analysis. Hermosilla et al., 2018 codified the skin colors of film actors by asking workers on Amazon Mechanical Turk to rate the skin tone on a five-point scale: “very light,” “light,” “medium,” “dark,” and “very dark.” Mitchell, 2020 recruited two coders to extract the skin tones of people on fashion magazine covers. The two coders were asked to first compare people’s skin tone with the Pantone skin Matching System (PMS) tone color strip ranging from PMS 726 to PMS 732, and then select the PMS category whose color is closest to the targeted skin tone as its measure. One could imagine it needs substantial hours to physically code human skin tone from hundreds of thousands of images available online for business insights. Moreover, the manual approach is economically unfeasible to process video messages providing the unstructured formats and inherent data diversity (Balducci & Marinova, 2018; Sheth & Kellstadt, 2021). The method is restricted by its sample sizes and potential biases of the diversity perceptions.

A recent study made the first attempt to conduct a visual computation analysis on skin tones used in mainstream clothing retail brands in the US (Butkowski et al., 2022). They demonstrated how such objective analysis of the representation of various skin tones could reveal the role of structural biases. However, the authors collected limited samples and used manual coders to crop human skin. We extend and automate the approach suggested by this study. Specifically, we propose a tool based on ML techniques to process large amounts of unstructured social media data to extract essential insights about business DEI signaling. To that end, we overcome two main methodological challenges: 1) automatically locating the skin of the people depicted in the visual content and subsequently quantifying their skin tone, and 2) objectively quantifying the diversity of visual marketing communications based on a series of imagery and videos.

Humans have the ability to recognize humans in an image and immediately process their skin color. However, this is a complicated process involving a

machine’s learning process before it is qualified to apprehend human skin. First, it is fundamental for a machine to determine what set of pixels represents human skin in an image. After that, the color of the pixels can be used to assess the skin tone. At last, by processing a set of images and videos, the diversity across visual marketing communications can be quantified. We deploy a supervised ML semantic segmentation to automate the extraction of skin pixels from visual content. Semantic segmentation is the process of automatically classifying each pixel in an image (or video frame) into semantic labels (i.e., books, birds, and cars) (Minaee et al., 2021). Along the semantic segmentation research stream, a series of models have been proposed, including traditional approaches (e.g., thresholding (Otsu, 1979)) and deep learning-based models (e.g., DeepLabV3 (Chen et al., 2017)). We refer readers to Minaee et al. (Minaee et al., 2021) for an extensive review. Because of the high accuracy, we use deep learning-based semantic segmentation models to extract skin pixels.

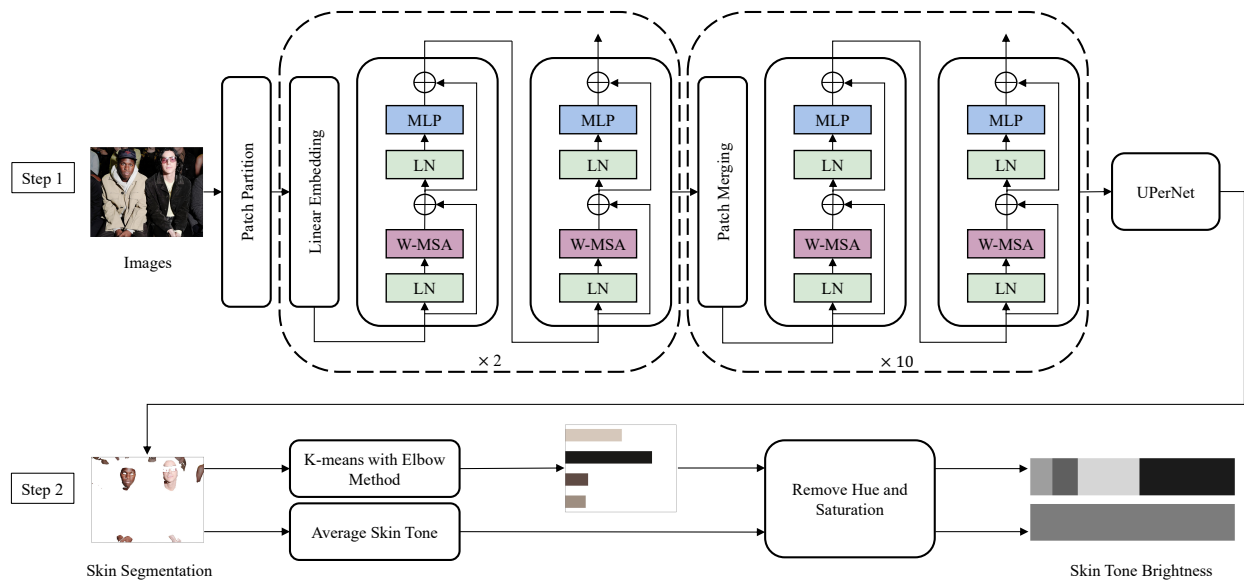
Once the segmentation model extracts skin pixels from the visual, we can obtain each skin pixel’s RGB (i.e., red, green, and blue components) values. We compute the average skin tone of people involved in an image by averaging the whole RGB values. In addition, we use unsupervised clustering methods to automatically classify skin pixels into different clusters to account for multiple people with distinct skin colors in an image. Following that, we can obtain the skin tone brightness of the image to show how dark or light the human color is. It is feasible to understand how diverse the skin tone representation is in brands’ visual communications over time with the skin tone measure.

3. Method

We illustrate the main framework of the proposed methodology to detect skin tone diversity in images and videos in Figure 1. In general, the methodology includes the following two steps. In Step 1, we deploy a supervised machine learning approach where an encoder-decoder model is trained to extract skin pixels from images. In Step 2, we compute the average skin tone of the extracted skin pixels, and employ K-means clustering to classify the extracted skin pixels into optimal clusters automatically. We describe the process of detecting the skin tone diversity measures in the following subsections.

3.1. Skin Pixel Extraction

It is necessary to recognize the human skin before quantifying the skin tone in an image. In this



MLP: multilayer perceptron; LN: layer norm; W-MSA (SW-MSA): multi-head self attention modules with regular (shifted) windows

Notes: we first train Swin Transformer with UPerNet decoder to segment skin pixels from a image in Step 1, followed by measuring skin tone and skin tone components in Step 2 accounting for multiple people with distinct skin colors in a single image.

Figure 1. Framework of the proposed methodology.

study, we select Swin Transformer and UPerNet as the encoder and decoder, respectively, to segment skin pixels from images because of the following three reasons (Liu et al., 2021; Xiao et al., 2018). First, draw on the Attention structure (Vaswani et al., 2017) in natural language processing, Swin Transformers can yield deep hierarchical feature maps which learn representations from images in various scales comprehensively. Because each representation is computed with its shifted window technique, Swin Transformer has a linear computational complexity proportional to its input image size, overcoming previous quadratic complexity. Second, UPerNet (Xiao et al., 2018), a unified perceptual parsing for multi tasks, succeeds in categorizing scenes and detecting objects along with identifying the textures and surfaces. Third, the performance of the Swin Transformer with UPerNet surpasses its previous state-of-the-art in semantic segmentation (Liu et al., 2021).

The proposed encoder-decoder model is a supervised ML approach. To train the skin segmentation model for extracting human skin pixels in images (or video frames), we used the skin and face detection (FSD) dataset (Phung et al., 2005) attributed to the following three reasons. First, the FSD dataset contains a large set of images (4,000 samples), and the annotated skin and face ground-truth are precise (Lumini & Nanni, 2020). Second, the dataset incorporates complete skin types (e.g., whitish, pinkish, yellowish, light brownish,

reddish, darkish, and dark brownish), which is essential in guaranteeing the detection accuracy of human skin of any color. Third, images in the dataset have different lighting conditions (e.g., indoor and outdoor). It further promises success in detecting skin pixels from images photographed in diverse contexts.

We implement the skin segmentation through the following three processes. First, we randomly split the FSD dataset into training and testing datasets with 3,200 and 800 images, respectively. Second, we constructed the encoder-decoder model with Swin Transformer and UPerNet using OpenMMLab¹ library in Python and trained the model. The intermediate trained models were validated after each training epoch and saved every five epochs for the best model selection. Finally, we selected the best checkpoint after comparing the validation accuracy and training loss. We assessed its performance on the testing dataset with the intersection over union (IOU) metric, which is widely used in semantic segmentation (Liu et al., 2021) and formulated as:

$$\text{IOU} = \frac{\text{Groud-truth Pixels} \cap \text{Segmented Pixels}}{\text{Groud-truth Pixels} \cup \text{Segmented Pixels}}$$

The IOU metric evaluates the accuracy of a trained model as the ratio of the intersection over the union between manually labeled skin pixels and

¹<https://github.com/open-mmlab/mmssegmentation>

model-detected skin pixels. In addition, we executed the baseline model of the thresholding approach (Naji et al., 2019) and applied the human skin color thresholds from Deepgaze² to extract skin pixels from the same 800 testing images. We report the performance comparison in the next section.

Brands post images with people to promote products. Some images contain human faces while others may not, such as an image incorporated with a bag in a person's hands but without his/her face included. To explore potential color differences between human faces and their whole skin, we trained another face segmentation model under the supervision of annotated face ground-truth in the FSD dataset. We used the face detection model³ to validate the segmentation accuracy because of its high accuracy. Rather than recognizing skin at the pixel level, face detection captures the face area at the image level. The detection results are informative of the number of people with faces in an image. With skin pixels accessible, we can now measure the skin tone.

3.2. Skin Tone Measure

The goal of measuring skin tone is to uncover how light or dark the human skin is. The skin pixels within one image might come from different parts of people (i.e., hands, arms, and face), and hence their colors vary from one to another. Therefore, one standard criterion is indispensable for comparing skin tone representations of people present in different images. Instead of only considering part of the skin or just the face, we first compute the average skin tone of fashion people inside an image by operations in the RGB (Red, Green, and Blue) color space, leading us to a single RGB color point, i.e., (45, 120, 98). Next, we convert the RGB color to HSV (hue, saturation, and value) color because HSV color space is closer to the way in which humans describe colors in nature (Naji et al., 2019). For example, one would refer to the color of the sky as dark blue or light blue instead of giving the percentage of mixing the three primary colors red, green, and blue.

In the HSV color space, hue (H) is a measure of the spectral composition of a color (e.g., red and yellow). Saturation (S) refers to the purity of a color, and value (V) reflects how light or dark a color is. Therefore, we remove the Hue and Saturation of the average skin tone of people in each image and obtain the brightness value ranging from 0 (black) to 255 (white) as the skin tone measure. The skin tone measure serving as a proxy of the color of the whole skin pixels is more comprehensive

than the prior approach used in Mitchell, 2020 where the author only took into consideration the skin patch between the top portion of the eyebrows and neglected other parts of human skin.

Moreover, we observed a considerable amount of images that include multiple people with distinguishable skin tones. The average skin tone of such images results from the trade-off among different skin colors. We further introduce skin tone components allowing for analyzing the percentage of varying skin tones, especially those existing significant distance (i.e., very dark and very light). To differentiate skin tones inside one image, we utilize the mini-batch K-mean algorithm (Sculley, 2010). The mini-batch K-mean model classifies each pixel into different clusters iteratively with random batches of image pixels rather than the full pixels (used in the K-means algorithm) to save memory and increase efficiency. During the classification, the optimal number of clusters is selected under the guidance of the Elbow method (Satopaa et al., 2011).

After obtaining the skin tone brightness of the whole skin and main components, we next propose skin tone diversity measures that offer straightforward views about different skin tone representations in images.

3.3. Skin Tone Diversity Measure

Existing research such as Mitchell, 2020 used categorical skin tone to show the skin tone diversity. In this study, we automate and extend the skin tone thresholds from Mitchell, 2020 to assign each brightness value to one of the following five exclusive categories:

$$\text{Category} = \begin{cases} \text{Very dark} & 0 < \text{brightness} \leq 98, \\ \text{Dark} & 98 < \text{brightness} \leq 138, \\ \text{Medium} & 138 < \text{brightness} \leq 193, \\ \text{Light} & 193 < \text{brightness} \leq 224, \\ \text{Very light} & 224 < \text{brightness} < 255. \end{cases}$$

Analogously, we can obtain the categorical measure for each skin tone component and its percentage (ratio of the number of pixels in one cluster over the number of whole image pixels) accordingly.

The categorical skin tone measure makes it practicable to view the historical representation of each skin tone category. However, it might suffer from information loss by categorization. For this reason, we include continuous skin tone diversity measures that interpret the skin tone diversity in a more granular way. We compute the variance and mean in monthly posted images as two continual diversity measures. Since brightness values 0 and 255 refer to black and white, respectively, a large (small) mean of the skin tone

²<https://github.com/mpatacchiola/deepgaze>

³Dlib face recognition built on deep learning has an accuracy of 99.38% on the Labeled Faces in the Wild benchmark

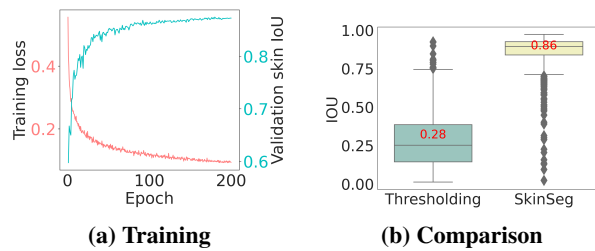


Figure 2. Performance of the skin segmentation model.

brightness values is interpreted as more representation of people with light (dark) skin tones. In contrast, a large (small) variance reflects a more diverse (monotonous) representation of skin tones.

3.4. Results

We first show how well the approach performed to convince our findings pertinent to the skin tone representation dynamics using the proposed skin segmentation approach (provided in the following subsection). With the 3,200 training images from the FSD dataset, we train the encoder-decoder model with Swin Transformer and UPerNet for 200 epochs on the training dataset. We set the initial learning rate as 0.0001 and batch size as 8. After completing the training process, we test the performance on the 800 out-of-sample images. Figure 2 provides the performance of the training process and the IOU comparison with the baseline in segmenting skin pixels. As illustrated in Figure 2a, the training loss curve decreases sharply during the beginning epochs and holds stable in the end while the validation mean IOU (the average of IOU over the whole 800 images) increases and converges, which confirms the success of the training process.

We next plot the IOU boxplots in Figure 2b to compare the performances between the trained skin segmentation model and the thresholding baseline. The boxplots show that the mean IOU of our proposed model is 0.86, while the mean IOU of the baseline is 0.28.

Using the trained segmentation models, we extracted skin and face pixels in images collected from the 34 fashion brands, described in more detail in the next section. Figure 3 illustrates three representative samples from the data. The first row shows the results of one image with two people that have distinct skin tones. Our trained skin segmentation model extracted the skin pixels of the two people and the skin pixels of people in the background. Compared to the face extraction results, the face detection results include a small portion

of extra pixels because face detection operates the image at the image level instead of pixel-level and surrounds the face area with a rectangle. However, it is acceptable in that we merely use face detection results to validate the accuracy (i.e., the number of faces in one image) of segmentation results. After the extraction, the K-means algorithm automatically clustered four skin tone components with different skin pixel percentages in the first sample image. The average skin tone brightness of the image and the skin tone brightness of various components are illustrated at the top of the last column in Figure 3. Similarly, our methodology works well in the second sample with one person and the third without any human face.

In the following section, we present an example application of using our tool to study the skin tone representation dynamics of 34 fashion brands.

4. Empirical Application: Fashion Brands' Skin Tone Dynamics

To illustrate the importance and effectiveness of our proposed method and resulting online tool, we will describe how it can be used to study diversity and representation in brands' visual marketing communications. In this example, we study fashion brands on social media platforms Instagram and Twitter. We chose to study fashion brands due to their prominent use of people in their marketing communication and ongoing discussion around the lack of diversity in the industry (Mitchell, 2020). We selected 34 fashion brands according to their ranks on the Lyst index⁴. The Lyst index is a quarterly ranking of fashion's "hottest" brands taking into account a list of comprehensive factors such as consumer searches, social media mentions, product views, and sales. We collected all their posts over two years, from 05/26/2019 to 05/26/2021, on Instagram and Twitter platforms. Table 1 records the number of gathered samples in each unstructured data type.

Source	Images	Videos	Texts	Gifs
Instagram	54,976	8,012	37,696	-
Twitter	19,865	5,858	26,242	123

Table 1. Social media posts.

We obtained fashion people's skin pixels and skin tones in all brand-generated images, excluding black and white visual samples, using our proposed methodology. In the following paragraphs, we first

⁴We selected brands that were repeatedly appeared in the Lyst index <https://www.lyst.com/data/the-lyst-index/q421/> and that are active on either of the two social media platforms

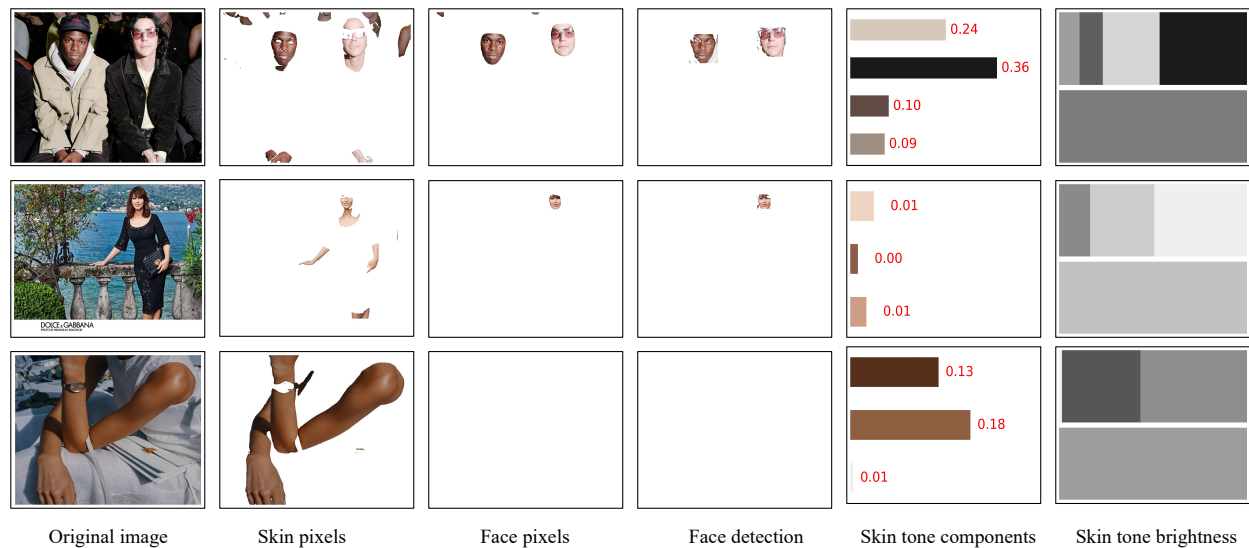


Figure 3. Results from the proposed methodology.

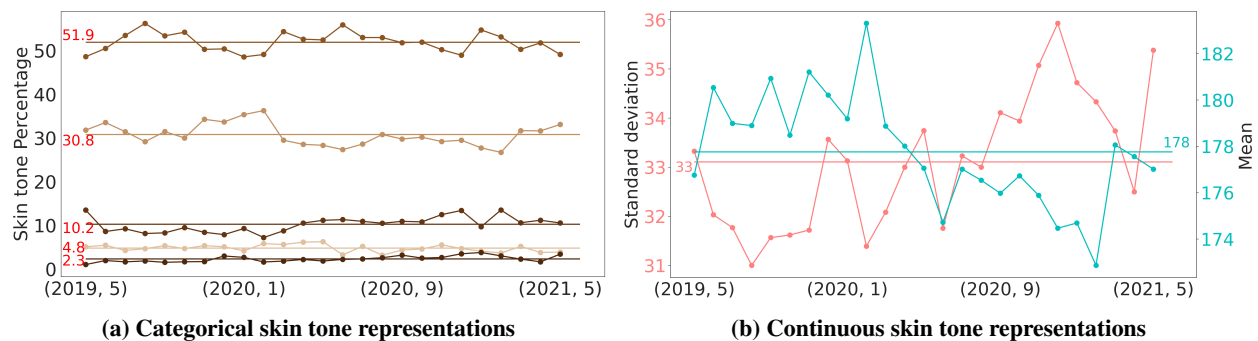


Figure 4. Skin tone diversity dynamics.

report the aggregated skin tone representation dynamics of the 34 fashion brands using the categorical and continuous skin tone diversity measures. We next document each individual brand's diversity position and the differences between luxury and commercial brands.

4.1. Aggregated Dynamics

We acquired the skin tone category of each image (i.e., dark, medium, or light) according to the brightness thresholds discussed in the prior section. To shed light on the historical variation of skin tone representations, we computed the monthly percentage of each skin tone category, that is, the ratio of the number of images in the category over the number of total images posted in the month. Figure 4a shows the monthly dynamics of the

five skin tone categories (i.e., very light, light, medium, dark, and very dark) where the color of each line is reflective of the corresponding skin tone category. In the long run, medium skin tone representations account for the most, with a mean percentage of 51.9% followed by light skin tone representations (30.8%) and dark skin tone representations (10.2%). The mean percentages of very light and very dark skin tone representations are 4.8% and 2.3%, respectively.

In addition, the results show that the percentage of medium skin tone representations fluctuates in a short time but is stable overall. Interestingly, the percentage of light (dark) skin tone representations varies and decreases (increases) starting in February 2020. The percentage of very light (very dark) skin tone changes slightly over months but tilts down (tilts

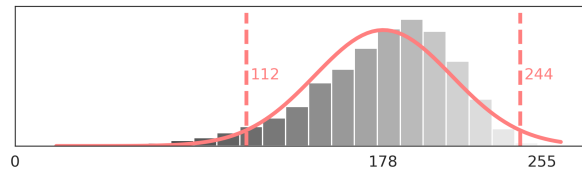


Figure 5. Two-sigma effect.

up) gradually from May 2020. Because the beauty standard (i.e., white, young, and skinny femininity) is deeply institutionalized (Mears, 2010) within the fashion industry, brands tend to change slowly and incrementally (Evans, 2014).

The categorical skin tone measure illustrates skin tone dynamics over time. However, to provide a holistic overview, we further report results from our proposed continuous skin tone measures in Figure 4b. Instead of assigning each image to one of the five skin tone categories, we calculated the variance and mean of skin tone brightness in monthly posted images. The variance and mean provide a reflection of the variation of fashion brands' communication signaling. As shown in Figure 4b, the long-term mean of monthly standard deviations (we plot standard deviation for interpretation simplicity) is 33, while the mean of monthly means is 178. In other words, there is a 95% probability (two-sigma effect) that the skin tone brightness tends to vary in the range of 112 ($178 - 2 * 33$) to 244 ($178 + 2 * 33$) as illustrated by the collective skin tone brightness histogram and fitted Gaussian distribution in Figure 5. Furthermore, the results show that the variance curve oscillates more frequently in a range before July 2020 and breaks the range to tilt up evidently afterward. In contrast, the mean curve swings and starts to break the range downwards from April 2020. It is noted that the skin tone brightness 0 and 255 refer to black and white, respectively. The downtrend of the monthly skin tone brightness mean curve suggests that fashion brands are increasing the representation of people with darker skin tones, which is affirmed by the uptrend of the monthly variance curve. For the illustration purpose, Figure 6 shows images with different average skin tone brightness to understand the magnitude of variance.

In the above paragraphs, we explored the skin tone representation dynamics of the 34 fashion brands in an aggregated approach. We next describe the results of the skin tone distribution of each brand in an individual manner.

4.2. Individual Dynamics

With the proposed continuous skin tone diversity measures, we first show the diversity position of typical and partial brands in Figure 7.⁵ The x-axis shows the average monthly skin tone brightness mean, and the y-axis shows the average monthly standard deviation in the skin tone brightness. The bar along the horizontal axis shows the skin tone brightness accordingly, and the square point with gray color shows the location of the industry standard. The scatter plot suggests that Rick Owens has the largest average skin tone brightness (189.6) while Fear of God has the smallest average skin tone brightness (150.0). In terms of the brightness standard deviation, Marine Serre has the largest value (35.2) while Stone Island has the smallest value (24.2). It is noted that our proposed diversity measure takes into account not only the brightness of skin tone representation but also the variance of skin tone brightness in monthly posted images. Hence, results in Figure 7 could offer hints for diversity comparisons among fashion brands and the industry standard.

In addition, we explore the difference between commercial and luxury brands. As illustrated in Figure 8, we plot the skin tone brightness histograms of representative commercial and luxury brands, where dashed lines indicate the two standard deviations of the mean. The results show that overall, left tails of the histograms are longer than right tails, suggesting darker skin tones are more likely to be fitted as outliers because of fewer representations. This phenomenon is more noticeable in luxury brands. Furthermore, the brightness bars across the skin tone spectrum are more evenly allocated in commercial brands than in luxury brands. The findings show that the representation of skin tone is more diverse in commercial brands, which could be understood that commercial brands tend to target their products to a more diverse group of consumers (Mears, 2010).

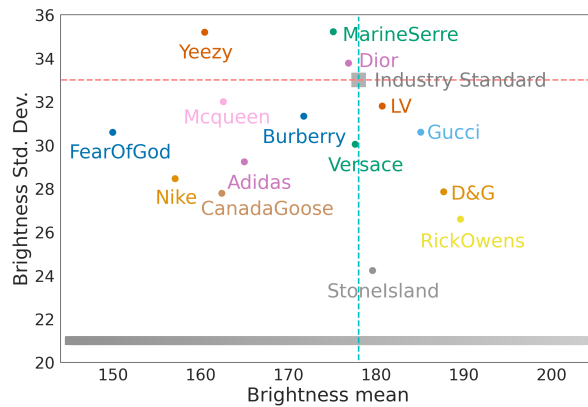
5. Discussion and Conclusion

Firms and researchers spend increasingly more time and resources on diversity, equity and inclusion. Firms are interested in improving DEI for a variety of reasons, including improving employee and customer satisfaction, enhancing perceived trust, gaining an edge over their competition, and due to the general pressure from society to do better. An important component in signaling DEI efforts and general DEI to society is the

⁵We exclude Balenciaga because it only posted 11 images on Twitter and Instagram over the two years, which is far less than other selected fashion brands.



Figure 6. Visual interpretation of variance where numbers below images present the average skin tone brightness.

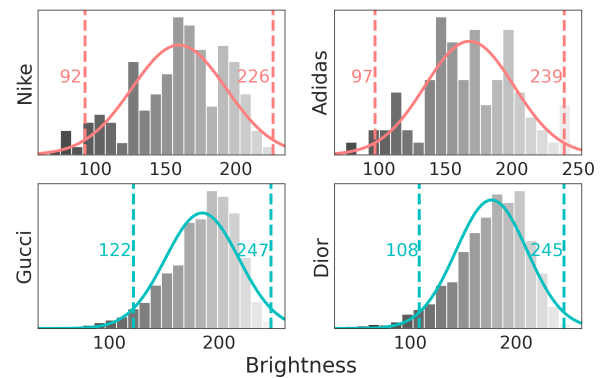


Notes: the gray square marks the location of the industry standard (178, 33). The horizontal bar along the x-axis illustrates the skin tone brightness change accordingly.

Figure 7. Brand diversity position.

use of strategic visual marketing communication. As a result, these visual communications provide insightful information about companies' strategies regarding DEI signaling. It is, therefore, important to be able to detect and quantify the diversity and representation of people within online visual content. At the same time, analyzing images and videos is a challenge for marketing research due to their unstructured nature (Balducci & Marinova, 2018). For these reasons, this study sought to propose an automated and objective framework to obtain structured diversity from unstructured data.

The framework relies on image mining and general machine learning techniques to detect and quantify skin tones in images and video. We first trained a skin segmentation model that was three times more accurate than the thresholding baseline method, highlighting its effectiveness in extracting skin pixels. We further proposed novel skin tone diversity measures based on the framework, including categorical and continuous approaches, which offer a comprehensive understanding of the skin tone dynamics of visual marketing communications at scale and over time.



Notes: the first row and second row show two commercial and luxury brands, respectively, where the vertical dashed lines present the two-sigma locations.

Figure 8. Typical brands' brightness distributions.

To illustrate the use of our proposed tool, we studied skin tone representations of firm-generated imagery on two major social media platforms for the 34 largest fashion brands. The categorical skin tone measure offers marketers and researchers a straightforward approach to understanding the diversity signaling of brands. Drawing on the skin tone thresholds, the five representation percentages can offer marketers an explicit hint about the deployment of people with different skin tones and the diversity signaling. Our proposed continuous diversity measures, the variance and mean of skin tone brightness, further provide a standard to compare the diversity positions of brands within the industry. Through the two-dimensional coordinate, the diversity position of each brand is precisely depicted. Hence, companies can be aware of their own and competitors' relative diversity positions in the industry.

In general, the findings presented in this study shed light on understanding the skin tone dynamics of people in visual marketing communications. Naturally, this study has limitations. First, we adopt the categorical skin tone thresholds from Mitchell, 2020 who obtained the thresholds from a survey with 249 respondents.

More people from diverse backgrounds could be recruited to increase the generalizability. Second, we selected 34 fashion brands from the Lyst index for this study. More data from other industries (i.e., cosmetics) could be collected to extend the robustness of our findings. Third, to provide a more holistic view and ensure that it encapsulates multiple definitions of diversity and inclusive representation seen in different fields, we intend to increase the number of ways diversity is measured and detected. Additionally, we intend to run a series of experiments to validate the relationship between what is measured and how people perceive it.

Our proposed tool that accurately and automatically analyzes skin tone in images and videos can offer many opportunities for researchers and practitioners. It helps overcome challenges related to processing large amounts of data and has the potential to provide a holistic view of diversity efforts communicated through visual contents. Therefore, we conclude by listing several fruitful areas for future researchers to explore using our framework. First, business managers can use the tool to monitor brands' DEI signaling and track diversity efforts across their communication ecosystem. Being aware of both their own and competitors' DEI positions, companies are able to adjust their visual marketing strategies dynamically to meet consumer needs and/or to change institutional norms. Second, because of the easy access to structured skin tone measures with our tool, it is feasible for marketing researchers to explore the potential impact of skin tone representation in visual media on business outcomes such as consumer attitudes, brand awareness, purchase intent, and even shareholder values. In summary, this study extends the boundary of investigating DEI in large unstructured data, and our tool allows researchers to include objective diversity metrics in the study of DEI broadly in marketing.

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